

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	08 March 2026
Team ID	
Project Name	Electric Vehicle Analytics Dashboard
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR1	Dataset Collection	Collect EV datasets such as EVIndia, ElectricCarData_Clean, and Charging Station datasets
FR-2	Data Storage	Store EV datasets into PostgreSQL database tables
FR-3	Data Processing	Clean and prepare data using Python
FR-4	Database Connection	Connect PostgreSQL database with Tableau
FR-5	Data Visualization	Create charts and graphs for EV analysis in Tableau
FR-6	Dashboard Creation	Build interactive dashboards for EV analytics
FR-7	Data Filtering	Apply filters such as Brand, PowerTrain, and Car Model
FR-8	Dashboard Story	Create Tableau story scenes explaining EV insights
FR-9	Web Integration	Embed Tableau dashboard into Flask web application
FR-10	Dashboard Access	Allow users to view and interact with EV dashboards

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should be easy to understand and user-friendly for all users
NFR-2	Security	Access to the PostgreSQL database should be protected using authentication
NFR-3	Reliability	The system should provide accurate EV analytics without data errors
NFR-4	Performance	Dashboards should load quickly and handle EV datasets efficiently
NFR-5	Availability	The dashboard should be accessible through the web interface anytime
NFR-6	Scalability	The system should support adding more EV datasets and visualizations in the future